UNIT 1 — Q1. Characteristics and Types of Social Media Analytics (SMA)

Social Media Analytics (SMA) is the process of collecting, measuring, analysing and interpreting data from social media platforms to gain useful information for decision-making. It helps organizations understand customers, monitor brands, identify trends and improve marketing.

Characteristics of SMA:
1. Data-based: It uses data such as posts, comments, likes, shares, views, clicks, hashtags and reviews.
2. Real-time monitoring: It can track conversations and reactions as they happen.
3. Customer-focused: It helps understand customer opinions, needs, preferences and satisfaction.
4. Measurable: Performance can be measured using reach, impressions, engagement rate, clicks and conversions.
5. Supports different types of data: It can analyse text, images, videos, emojis, links and user interactions.
6. Trend and sentiment detection: It helps identify popular topics and whether opinions are positive, negative or neutral.
7. Decision support: The findings help improve content, advertising, customer service and products.
8. Continuous improvement: Results are monitored so that strategies can be changed when needed.

Types of SMA:
1. Descriptive analytics: Explains what happened, such as the number of likes or views received.
2. Diagnostic analytics: Explains why something happened, such as why one post received more engagement.
3. Predictive analytics: Uses past data to estimate future trends, engagement or demand.
4. Prescriptive analytics: Suggests actions to improve results, such as changing the posting time.
5. Sentiment analytics: Identifies positive, negative or neutral opinions.
6. Text analytics: Extracts keywords, topics and meaning from comments and posts.
7. Network analytics: Studies connections among users and identifies communities or influencers.
8. Action and content analytics: Measures user actions and compares the performance of different content formats.

Example: A company analyses comments, shares and views after launching a product. It discovers customer concerns and popular features, then improves its marketing and product decisions. Thus, SMA turns social media data into useful insights.

UNIT 1 — Q2. SMA in Small and Large Organizations

Social Media Analytics (SMA) helps both small and large organizations understand their audience and improve their marketing. However, their budgets, data volumes, tools and objectives are different.

SMA in Small Organizations:
Small organizations usually have limited budgets, small teams, fewer customers and local or regional markets. Their main objectives are to increase local brand awareness, build customer relationships, promote products at low cost, improve customer service and generate leads or sales. They commonly use Instagram, Facebook, WhatsApp Business, Google Business Profile and LinkedIn.

Important metrics include likes, comments, shares, reach, followers, engagement rate, website clicks and customer reviews. Tools may include Meta Business Suite, Google Analytics, Buffer, Canva Insights and Zoho Social. These tools help small businesses identify popular posts, understand customer preferences and respond quickly to feedback.

Benefits include affordable promotion, quick customer feedback and better engagement. Challenges include limited funds, lack of skilled analysts, small datasets, time constraints and difficulty measuring return on investment (ROI).

Example: A local bakery checks Instagram engagement and orders to learn which cake designs are popular. It then promotes similar cakes during festivals.

SMA in Large Organizations:
Large organizations often operate nationally or globally and generate huge amounts of data from platforms such as Facebook, Instagram, LinkedIn, YouTube, X, TikTok, blogs, reviews and discussion forums. Their objectives include global brand management, market intelligence, customer sentiment analysis, competitor monitoring, crisis management and product improvement.

They may use Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), image analysis, predictive analytics and network analysis. Tools include Sprinklr, Brandwatch, Meltwater, Talkwalker, Salesforce Marketing Cloud and Adobe Analytics. Important metrics include brand sentiment, share of voice, customer lifetime value, conversion rate, campaign ROI, engagement and viral reach.

Large organizations benefit from strategic planning, improved marketing and stronger customer loyalty. Their challenges include managing large datasets, privacy compliance, integrating data sources, fake accounts, high costs and data security.

In conclusion, small organizations generally use simple and affordable analytics for local engagement and sales, while large organizations use advanced tools and techniques for large-scale and strategic decisions.

UNIT 1 — Q3. Seven Layers of Social Media Analytics

The seven layers of Social Media Analytics explain how social media data moves from its source to useful business decisions.

1. Data Sources Layer: This is the foundation. Data comes from Facebook, Instagram, X, LinkedIn, YouTube, TikTok, Reddit, blogs, forums and online reviews. Data includes posts, comments, images, videos, likes, shares, hashtags and profiles.

2. Data Collection Layer: Data is gathered from different platforms using APIs, web scraping, RSS feeds and monitoring tools. Examples of tools are Hootsuite, Brandwatch, Sprinklr, Meltwater and Meta Business Suite. This layer supports continuous data collection.

3. Data Storage Layer: Collected data is stored for future processing and analysis. Storage options include relational databases, NoSQL databases, data warehouses, data lakes and cloud storage. Examples include MySQL, MongoDB and Amazon S3.

4. Data Processing Layer: Raw data is cleaned and organized. Duplicate records, spam and irrelevant information are removed. Text may be tokenized, stop words removed and languages identified. This improves data quality.

5. Data Analysis Layer: Statistical methods, AI, machine learning and NLP are used to extract insights. Techniques include sentiment analysis, text mining, trend analysis, topic modelling, network analysis and predictive analytics.

6. Visualization and Reporting Layer: Results are shown through bar charts, pie charts, line graphs, heat maps, dashboards, word clouds and network graphs. Tools include Excel, Tableau, Power BI and Google Looker Studio. Visualization makes results easier to understand.

7. Decision-Making and Action Layer: Organizations use the findings to choose target audiences, improve content, adjust advertising budgets, handle complaints, select influencers and optimize posting schedules. The results are monitored to improve future decisions.

Example: A company collects comments about a new phone, removes spam, analyses customer sentiment, displays the results on a dashboard and improves the phone based on common complaints.

Thus, the layers help transform raw social media data into meaningful insights and practical actions.

UNIT 1 — Q4. Explain the SMA Cycle

The Social Media Analytics (SMA) cycle is a continuous process of setting goals, collecting data, analysing it and using the results to improve social media performance. It helps organizations make decisions based on evidence instead of guesses.

1. Define Business Objectives: The first step is to decide what the organization wants to achieve. Objectives may include increasing brand awareness, improving customer satisfaction, measuring a campaign or generating sales leads. Clear objectives guide the choice of metrics.

2. Data Collection: Data is collected from platforms such as Instagram, Facebook, YouTube, LinkedIn, X, blogs and review websites. It may include posts, comments, likes, shares, clicks, views, hashtags and profiles. APIs and analytics tools can be used.

3. Data Preparation or Cleaning: The collected data is cleaned before analysis. Duplicate records, spam, fake accounts and irrelevant content are removed. Text is organized and prepared so that the results are more accurate.

4. Data Analysis: The cleaned data is examined using descriptive, diagnostic, predictive and prescriptive analytics. Techniques may include sentiment analysis, trend analysis, text mining, network analysis and influencer analysis.

5. Visualization and Reporting: Results are presented through charts, graphs, heat maps and dashboards. This makes patterns easier to understand and communicate.

6. Decision-Making: The organization uses the findings to improve its strategy. For example, it may change content topics, target a different audience, adjust its advertising budget or respond to complaints.

7. Implementation: The selected changes are applied by publishing improved content, launching campaigns, working with influencers or updating promotional strategies.

8. Monitoring and Feedback: Performance is checked using engagement rate, reach, impressions, click-through rate, conversions, customer sentiment and ROI. The organization compares results with its original objectives.

Example: A company wants to increase Instagram engagement. It studies previous posts, discovers that short videos perform better, publishes more videos and checks whether engagement improves.

The cycle repeats because audience behaviour and trends change. Continuous monitoring helps organizations learn from results and improve future campaigns.

UNIT 1 — Q5. Describe Search Engine Analytics

Search Engine Analytics is the process of collecting and analysing data about search queries, website visibility, rankings, traffic, users and conversions. It helps organizations understand how people find a website through search engines and how its search performance can be improved. It is an important part of Search Engine Optimization (SEO) and digital marketing.

Objectives:
1. Understand what users search for.
2. Identify effective keywords and search intent.
3. Measure website visibility in search results.
4. Monitor traffic received from search engines.
5. Evaluate SEO performance and improve rankings.
6. Identify opportunities for new content.
7. Measure conversions generated through search.
8. Understand what users do after visiting a website.

Important Metrics:
1. Impressions: The number of times a webpage appears in search results.
2. Clicks: The number of times users click a search result and visit the website.
3. Click-Through Rate (CTR): The percentage of impressions that lead to clicks. CTR = (Clicks ÷ Impressions) × 100.
4. Average Position: The average ranking of a webpage in search results.
5. Keywords: Words or phrases entered by users into a search engine.
6. Organic Traffic: Visitors who arrive through unpaid search results.
7. Bounce Rate or Engagement: Helps show whether visitors interact with the website or leave quickly.
8. Conversion Rate: The percentage of visitors who complete an action such as purchasing a product or filling a form. Conversion Rate = (Conversions ÷ Total Visitors) × 100.

Tools:
Google Search Console reports search queries, impressions, clicks, CTR, position and indexing. Google Analytics measures traffic sources, user behaviour, engagement and conversions. Bing Webmaster Tools provides search performance and website inspection for Bing. SEO platforms also support keyword research, ranking tracking, backlink analysis and site audits.

Example: An online store notices that a product page receives many impressions but few clicks. It improves the page title and description, then checks whether CTR increases.

In conclusion, search engine analytics helps improve website visibility, attract relevant visitors and measure the success of SEO activities.

UNIT 2 — Q1. Explain Nodes, Edges and Tie with Example

Social Network Analysis studies the relationships and interactions between people, organizations or accounts. A social network can be represented as a graph made up of nodes and edges. The basic idea is: Social Network = Nodes + Relationships.

1. Nodes: Nodes, also called vertices, represent the entities in a network. A node may be a person, student, employee, company, website, social media account or group. For example, in a college friendship network, Rahul, Priya, Amit and Neha are four nodes.

2. Edges: Edges, also called links, represent relationships or interactions between two nodes. Examples include friendship, following, messaging, sharing, collaboration and mentioning. If Rahul and Priya are friends, a line connecting Rahul and Priya represents an edge. In a directed network, an arrow shows direction. For example, Rahul → Priya may mean Rahul follows Priya, but Priya may not follow Rahul.

3. Tie: A tie is the relationship or connection between two nodes. It may be strong or weak depending on factors such as the frequency of interaction, closeness and shared experiences. A strong tie may exist between close friends who communicate daily. A weak tie may exist between two professional contacts who interact occasionally. Weak ties can still be useful because they may provide new information or opportunities.

Example:
Suppose five students—A, B, C, D and E—are connected as follows:
• A is friends with B.
• A is friends with C.
• B is friends with C.
• C is friends with D.
• D is friends with E.

Here, A, B, C, D and E are nodes. The friendship lines are edges. Each friendship relationship is also a tie. This network can be studied to identify who has the most connections and how information might travel between students.

Importance:
Nodes, edges and ties help researchers understand network structure, identify influential users, find communities and study how information spreads. In social media marketing, a company can identify highly connected users and use them to spread information about a product.

Therefore, nodes represent entities, edges represent links, and ties describe the relationships connecting the entities.

UNIT 2 — Q2. Degree, Distribution, Density, Trust and Connectivity

These terms are important measures used in Social Network Analysis to understand how people or accounts are connected.

1. Degree: Degree is the number of direct connections or edges attached to a node. For example, if student A has five friends, A's degree is 5. In a directed network, in-degree counts incoming links and out-degree counts outgoing links. If 500 people follow an Instagram account, it has an in-degree of 500.

2. Degree Distribution: Degree distribution shows how many nodes have each degree value. It helps explain whether most users have few connections or whether some users have many connections. For example, most social media users may have a moderate number of connections, while a few influencers have millions of followers. These highly connected nodes are called hubs.

3. Density: Network density measures the number of actual connections compared with the maximum possible number of connections. A close group where nearly everyone interacts with everyone has high density. A network where users follow many accounts but rarely connect with one another has low density. Density helps explain how closely connected a group is.

4. Trust: Trust is the belief that another person, account or organization is reliable. It can develop through reputation, expertise, consistent information, positive experiences and repeated interactions. For example, customers may trust an online seller who delivers products on time, provides accurate descriptions and responds professionally. Trust can influence recommendations and online purchasing decisions.

5. Connectivity: Connectivity describes whether and how easily nodes can reach one another through direct or indirect paths. In a connected network, every node can reach every other node through some path. In a disconnected network, separate groups have no path between them. For example, A—B—C—D is connected, while A—B—C and D—E are separate components.

Importance:
These measures help organizations identify influencers, understand communities, study information spread and detect isolated groups. Degree helps identify popular accounts, density shows how closely a community interacts, trust helps explain reliable relationships, and connectivity shows whether information can travel through the network.

Together, these concepts provide a clearer picture of the structure and behaviour of social networks.

UNIT 2 — Q3. Sources of Location Data and Categories of Location Analytics

Location analytics combines geographical information with other data to understand where events, customers or activities occur. It is useful for business planning, marketing, navigation and decision-making.

Sources of Location Data:
1. Latitude and Longitude: These coordinates identify a precise position on the Earth's surface. Latitude measures distance north or south of the equator, while longitude measures distance east or west of the Prime Meridian. They are used in maps and navigation.
2. GPS-Based Data: The Global Positioning System uses satellites to provide location and time information. GPS is commonly used in smartphones, vehicles and navigation devices and can provide highly accurate locations.
3. IP-Based Data: IP geolocation estimates a device's location using its internet protocol address. It is used in web analytics, targeted advertising and fraud detection. It is generally less accurate than GPS.
4. Postal Address: A postal address includes details such as house number, street, city, state and postal code. Geocoding can convert an address into latitude and longitude coordinates.

Categories of Location Analytics:
1. Geographic Information Systems (GIS): GIS stores, checks, analyses and displays data connected to places on Earth. Example: A city uses a map of roads and houses to plan a new park.
2. Spatial Analytics: It studies how locations and objects relate to one another based on distance or position. Example: A retailer finds a suitable location for a new shop by studying customer locations.
3. Time Series Analysis: It tracks how location data changes over time. Example: Traffic authorities monitor vehicle speeds hourly to identify rush-hour periods.
4. Mapping: It displays data visually on maps so that patterns can be understood easily. Example: A weather app displays rainfall on a map.
5. Localized Intelligence: It uses local information to make decisions for a specific area. Example: A coffee shop sends a coupon to nearby customers.

Location analytics can help businesses identify customer areas, choose store locations, target advertisements and allocate resources. However, location data must be handled responsibly because detailed location histories can create privacy and safety risks. Organizations should be transparent and protect users' data.

UNIT 2 — Q4. Write a Note on Search Engine Tools

Search engine tools help website owners and digital marketers understand how websites appear in search results, what users search for and how visitors behave after clicking a result. These tools are useful for Search Engine Optimization (SEO), website improvement and digital marketing.

1. Google Search Console: This tool provides information about a website's performance in Google Search. It reports search queries, impressions, clicks, click-through rate (CTR), average search position and indexing issues. It helps website owners identify pages that need improvement and check whether pages can appear in Google Search.

2. Google Analytics: Google Analytics measures what happens after users reach a website. It provides information about traffic sources, user behaviour, engagement and conversions. For example, a business can find out how many visitors came from organic search and how many completed a purchase or filled out a form.

3. Bing Webmaster Tools: This tool helps website owners check how their websites perform in Bing search. It provides search-performance information and website inspection features. It can help identify issues that affect a website's visibility on Bing.

4. SEO Analytics Tools: Specialized SEO platforms provide additional features such as keyword research, ranking tracking, backlink analysis, competitor analysis, website audits and search visibility reports. They help marketers understand which keywords are useful and compare website performance with competitors.

5. Keyword Research Tools: These tools help identify the words and phrases people use when searching online. They support content planning and help businesses create pages that match users' search intent.

Uses and Benefits:
• Measure website visibility and search traffic.
• Find effective keywords and content opportunities.
• Monitor search rankings and clicks.
• Identify website or indexing problems.
• Understand visitor behaviour and conversions.
• Compare performance and improve SEO strategies.

Example: A college website uses Google Search Console to discover that students search for “TYDS admission details” but rarely click its page. The college improves its page title and description, then checks whether clicks increase.

In conclusion, search engine tools help websites attract relevant visitors, improve search visibility and measure the results of SEO efforts.

UNIT 2 — Q5. Write a Note on Network Visualization

Network visualization is the graphical representation of a network. It displays entities and their relationships so that the structure of a social network can be understood easily.

Main Elements:
1. Nodes or Vertices: These represent entities such as people, organizations, social media accounts, websites or groups.
2. Edges, Links or Ties: These represent relationships between nodes, such as friendship, following, messaging, sharing or collaboration.
3. Node Attributes: These are properties of nodes, such as user type, location, degree or community membership. They may be shown using size, colour or labels.
4. Edge Attributes: These describe connections, such as direction, interaction frequency or relationship strength. Arrows can show direction, while thicker lines may represent stronger or more frequent relationships.

A social network can be represented as a graph, G = (V, E), where V is the set of nodes and E is the set of edges. For example, on Instagram, each account is a node and a following relationship is an edge.

Uses of Network Visualization:
• Identify highly connected or influential users.
• Find communities or groups with common interests.
• Identify users who act as bridges between groups.
• Understand whether a network is densely or sparsely connected.
• Study how information, news or misinformation may spread.
• Detect isolated users or separate network components.
• Support influencer marketing and recommendation systems.

Tools:
Gephi is an open-source tool for creating network graphs, identifying communities and calculating centrality. NodeXL works with Microsoft Excel and helps import network data, calculate network measures and create visualizations. UCINET supports social network analysis, centrality and matrix analysis. Pajek is designed especially for analysing large networks. NetworkX is a Python library for creating and analysing networks.

Example: A college creates a graph of students who communicate with each other. The visualization shows one group of closely connected students and another student who connects two groups. The college can identify communities and understand communication patterns.

Thus, network visualization converts complex relationships into a visual form, making network analysis easier to interpret and use for decision-making.

UNIT 3 — Q1. Explain Various Network Types in Social Media

Social media networks can be classified according to their purpose and the way their users are connected. Understanding these types helps analysts study relationships, communication and information sharing.

Types Based on Purpose:
1. Personal Social Networks: These focus on connecting individuals, sharing updates and maintaining friendships. Examples include Facebook and Instagram.
2. Professional Networks: These support career development, recruitment, professional networking and knowledge sharing. LinkedIn is an example.
3. Content-Sharing Networks: These focus on sharing photos, videos or other media. Examples include YouTube, Instagram and Pinterest.
4. Discussion Networks: These allow users to ask questions, share opinions and take part in discussions. Examples include Reddit and Quora.
5. Messaging Networks: These support direct communication, voice and video calls, group chats and file sharing. Examples include WhatsApp, Telegram and Messenger.
6. Microblogging Networks: These allow users to publish short posts, replies, reposts, hashtags and mentions. X is an example.
7. Academic and Research Networks: These help researchers share publications, find collaborators and discuss research. Examples include ResearchGate and Academia.edu.

Types Based on Connections:
1. Directed Network: Relationships have a direction. For example, A $\rightarrow$ B may mean A follows B; B does not necessarily follow A.
2. Undirected Network: Relationships have no direction and work both ways. A mutual friendship can be shown as A—B.
3. Weighted Network: Connections have numerical values representing frequency or strength. If A and B exchange 100 messages, the edge may have a weight of 100.
4. Unweighted Network: An edge only indicates whether a relationship exists; it does not show frequency or strength.
5. Bipartite Network: It contains two different types of nodes, with links generally connecting one type to the other. For example, users can be connected to hashtags they use.

Example: A brand may study a directed follower network to identify accounts followed by many users, or a bipartite user-hashtag network to discover which topics are popular.

These network types help social media analysts identify influential users, understand communities, study engagement and examine how information spreads.

UNIT 3 — Q2. Discuss Network Analytics Tools

Network analytics tools are software applications used to study relationships among people, organizations, websites or social media accounts. They help analyse network structure, identify influential users, find communities and understand how information spreads.

1. Gephi: Gephi is an open-source network visualization and analysis tool. It creates network graphs, identifies communities, calculates centrality and displays large networks. It is useful for understanding complex relationships visually. Example: A college can use Gephi to visualize communication among students.

2. NodeXL: NodeXL works with Microsoft Excel. It can import network data, analyse social media networks, calculate network metrics and create visualizations. Its spreadsheet-based interface makes it useful for beginners and users familiar with Excel.

3. UCINET: UCINET is software designed for Social Network Analysis. It supports network analysis, centrality calculations, matrix analysis and network visualization. It is useful for examining relationships and the position of nodes within a network.

4. Pajek: Pajek is a network analysis and visualization program designed particularly for large networks. It can analyse social networks, communication networks, citation networks and organizational networks.

5. NetworkX: NetworkX is a Python library used to create, manipulate and analyse networks. It allows users to represent nodes and edges and calculate network properties. It is useful for people who want to automate analysis or integrate it into Python data-science projects.

Applications of Network Analytics Tools:
• Identify influencers and highly connected users.
• Find communities with common interests.
• Calculate degree and centrality.
• Study connections and shortest paths.
• Analyse the spread of posts, trends and news.
• Detect unusual relationships that may indicate fraud or suspicious activity.
• Examine collaboration in business, education and research.

Example: A company collects public interaction data and uses a network tool to identify users who connect different communities. These users may help a campaign reach different audience groups.

In conclusion, network analytics tools convert relationship data into measurable and visual information. The appropriate tool depends on the size of the network, the type of analysis required and the user's technical skills.

UNIT 3 — Q3. Write a Note on Social Media Text Analytics

Social Media Text Analytics is the process of extracting useful information from unstructured text on social platforms. The text may include posts, comments, tweets, reviews, replies, captions and discussion threads. It helps organizations understand customer opinions, identify trends and improve products or marketing.

Characteristics of Social Media Text:
1. Short Length: Posts and comments often contain only a few words.
2. Time Sensitivity: Trends and opinions may change within hours or days.
3. Informal Language: Users often write slang, abbreviations, emojis and non-standard grammar.
4. Large Volume: Social media generates large amounts of user-created text in real time.

Main Steps and Techniques:
1. Text Preprocessing: Text is cleaned before analysis. Tokenization splits sentences into words or tokens. Stop-word removal filters common words such as “the” and “is.” Stemming and lemmatization reduce words to their base forms.
2. Text Representation: Bag of Words represents text through the words it contains. TF-IDF estimates how important a word is within a collection of documents. N-grams examine sequences of words or other text units.
3. Sentiment Analysis: It identifies whether an opinion is positive, negative or neutral. For example, “The camera is excellent” is positive, while “The battery is poor” is negative.
4. Topic Modelling: It discovers common themes in a large collection of posts.
5. Entity Recognition: It identifies names of people, organizations and locations.

Tools:
Lexalytics supports semantic and sentiment analysis. Netlytic provides cloud-based text and social network analysis. LIWC analyses emotional and language patterns. DiscoverText supports collecting, cleaning and analysing text streams.

Applications:
• Monitor brand reputation.
• Understand customer satisfaction and complaints.
• Identify popular topics and emerging trends.
• Analyse product reviews.
• Support crisis management and content planning.

Example: A mobile company analyses thousands of comments after launching a phone. It finds that customers like the camera but complain about battery life. The company can use this information to improve customer service and future products.

Therefore, text analytics turns large amounts of informal social media language into useful insights for decision-making.

UNIT 3 — Q4. Describe Social Media Action Analytics

Social Media Action Analytics focuses on collecting and analysing the actions users take on social media rather than only the words they write. It measures behaviour, engagement and user interest to understand content performance and improve digital strategies. Text analytics studies what people say, while action analytics studies what people do.

Main Categories of User Actions:

1. High-Volume, Low-Effort Actions:
These actions require little effort and often show basic awareness or interest.
• Likes and reactions show appreciation or an emotional response.
• Views and impressions measure content visibility.
• Link clicks show that a user wants to explore more information.

2. High-Value Actions:
These actions require more active interest and may indicate stronger engagement.
• Shares, reposts and retweets spread content to a user's network and may increase its reach.
• Saves and bookmarks show that content may be useful for future reference.
• Follows and subscriptions show that a user wants to receive future updates.

3. Conversion Actions:
These are actions connected directly to organizational goals.
• App downloads.
• Lead generation through forms or registrations.
• Newsletter subscriptions.
• Purchases completed after seeing a social media campaign.

Important Metrics:
Action analytics may measure likes, comments, shares, saves, clicks, views, follows, sign-ups, purchases, engagement rate and conversions. These metrics help evaluate content popularity and campaign effectiveness.

Tools:
Hootsuite tracks engagement across platforms and can help identify effective posting times. Buffer provides easy-to-use scheduling and engagement analytics. Sprout Social offers charts, reporting and team collaboration. Socialinsider supports content performance and competitor comparisons. Meta Business Suite measures reach, impressions, likes, comments and shares. Google Analytics can track website visits and conversions from social media links, including links using UTM parameters.

Example: A business posts a product video. Many users view it, some click the website link, and a smaller number purchase the product. Action analytics helps the business understand each stage and identify where the campaign can be improved.

In conclusion, action analytics connects social media activity with audience behaviour and business outcomes. It helps organizations improve content, engagement and conversion rates.

UNIT 3 — Q5. Write a Note on Social Media Hyperlink Analytics

Social Media Hyperlink Analytics is the process of extracting, analysing and interpreting hyperlinks or URLs shared across social media platforms and websites. It helps organizations understand where users go after clicking a link, which platforms send traffic and which sources influence website visits.

Action analytics measures what users do within a social platform, such as liking or sharing a post. Hyperlink analytics focuses on links and what happens when users move between websites or platforms.

Types of Hyperlinks:
1. In-links or Inbound Links: These are links from external websites or social media platforms that lead to a particular website or profile. Example: A Facebook post contains a link to a company's landing page and sends visitors to it.
2. Out-links or Outbound Links: These are links on a website or page that direct users to another external website or platform. Example: A company blog contains a link to its Instagram profile.
3. Co-links: These occur when two separate websites both link to the same third-party source. Co-link analysis can help identify related websites, shared topics and connections between sources.

Uses of Hyperlink Analytics:
• Identify which social platforms send visitors to a website.
• Track clicks on links shared in posts and campaigns.
• Understand user journeys after a click.
• Measure traffic and conversions generated by each campaign.
• Study backlinks and the authority of websites linking to a page.
• Identify useful content sources and improve marketing strategies.

Tools:
1. Bitly: Shortens links and tracks clicks, approximate geographic information, devices and timestamps.
2. Google Analytics 4 (GA4): Uses campaign parameters such as UTM tags to identify which social network, campaign or advertisement generated website traffic and conversions.
3. Ahrefs and Moz: Analyse backlinks and the link environment of websites.
4. Hootsuite and Buffer: Help manage social posts and may provide link-shortening or click-tracking features.

Example: A company shares the same product link on Instagram, Facebook and LinkedIn. Using UTM parameters and Google Analytics, it compares traffic and purchases from each platform. It can then focus its efforts on the platforms that produce better results.

In conclusion, hyperlink analytics helps trace the movement of users across links, measure traffic sources and understand the contribution of social media to website visits and conversions.